Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/16/2018
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
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Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	Normal gravity flame blowoff limits in an axisymmetric polymethylmethacrylate (PMMA) rod geometry in upward axial stagnation flow are compared with microgravity Burning and Suppression of Solids – II (BASS-II) results recently obtained aboard the International Space Station (ISS). This testing utilized the same BASS-II concurrent rod geometry, but was done in normal gravity mixed convective flow. Cast PMMA rods of diameters ranging from 0.635 cm to 3.81 cm were burned at oxygen concentrations ranging from 15% to 18% by volume. The forced flow velocity where blowoff occurrs was determined for each rod size and oxygen concentration. These blowoff limits compare favorably with the BASS-II results when the buoyant stretch rate is included and the flow is corrected by considering the blockage factor of the fuel. From these results, the normal gravity blowoff boundary for this axisymmetric rod geometry is determined to be linear, with oxygen concentration and flow velocity as coordinates. We describe a new normal gravity ‘upward blowoff test’ method which extrapolates this linear blowoff boundary to the zero mixed stretch rate limit to resolve microgravity flammability limits—something current methods cannot do. This new test method can improve spacecraft fire safety for future exploration missions by providing a tractable way to obtain accurate but conservative estimates of material flammability in low gravity based on tests performed on Earth. During the blowoff extinction of clear cast PMMA rods in concurrent axial flow for microgravity BASS-II experiments, a dynamic flame oscillation was observed after the flame was blown off of the stagnation point but briefly stabilized on the periphery of the rod. Complementary normal gravity experiments were conducted and flame oscillations were tracked using a high speed color camera at 240 frames per second. The side-stabilized flame oscillated up and down the rod with increasing amplitude until the entire flame extinguished. In none of the BASS-II or normal gravity tests could the side-stabilized flames persist (Hopf subcritical bifurcation). Since the oscillations occurred even in microgravity, the mechanism does not depend on gravity. For the larger fuel radius tests, the flame developed asymmetric oscillation (pitchfork bifurcation). The oscillation time and the number of oscillations scale with the inverse square of the rod radius (~ Fourier no.) for the preheated microgravity rods. The average flame oscillation frequency is found to be linearly dependent on the mixed convective stretch rate (inverse of the flow time). The flame intensity varied in concert with its direction, either increasing or decreasing as the flame moved upstream or downstream, respectively. The oscillation frequency decreased as the amplitude increased and the flame slipped slightly farther down the rod with each oscillation. The flame speed increased with each subsequent oscillation, both flashing forward upstream and retreating downstream. The oscillations were found to closely follow a power law log-periodic dependence similar to those that describe systems approaching a critical point, such as diffusion-limited aggregation clusters, earthquakes, ruptures, and even stock market crashes. The net flame speeds varied linearly with ambient oxygen concentration, and linearly with the mixed convective stretch rate. Based on these observations, a mechanistic theory of the oscillations is described, and is consistent with the thermodiffusive instability. Low-pressure blowoff experiments were also conducted with a stagnation flame stabilized on the forward tip of cast PMMA rods in a vertical wind tunnel. Pressure, forced flow velocity, gravity, and ambient oxygen concentration were varied. Most of the tests were performed in normal gravity, but a handful of microgravity tests were also conducted to determine the influence of buoyant flow velocity on the blowoff limits. A linear shift of a buoyant flow of ~25 cm/s was adequate to match the microgravity data. The flame standoff distance before blowoff varies with the inverse square root of pressure via the gas phase length scale d=(a/a)1/2, where the thermal diffusivity varies inversely with pressure. The blowoff limits have a linear dependence between the partial pressure of oxygen and the total pressure, regardless of forced flow velocity and gravity level. Phenomenologically, the reaction rate is proportional to the oxygen flux to the reaction zone, so the flame surface area increases (via increasing d) as pressure is reduced, which compensates for the reduced partial pressure of oxygen for a given flow velocity. The flame standoff distance is also proportional to the inverse square root of the stretch rate (velocity), so the flame area decreases linearly with the mixed flow velocity. This reduced flame area results in a linear increase in the blowoff pressure so that the oxidizer molar flux to the smaller area

	maintained. The flow velocity (forced and/or buoyant) affects the blowoff pressure through the critical Damkohler number residence time, which dictates the partial pressure of oxygen at blowoff. This is because the critical stretch rate increases linearly with increasing pressure at low pressure (sub-atmospheric pressures) since a second-order overall reaction rate with two-body reactions dominates in this pressure range. A normal gravity materials flammability screening methodology is suggested for spacecraft exploration atmospheres.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Olson S, Ferkul P, Marcum J. "Analysis of High Speed Video of PMMA Rod Blowoff." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Articles in Peer-reviewed Journals	Olson S, Ferkul P. "Microgravity flammability boundary for PMMA rods in axial stagnation flow: Experimental results and energy balance analyses." Combustion and Flame. 2017 Jun;180:217-29. https://doi.org/10.1016/j.combustflame.2017.03.001 , Jun-2017
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Papers from Meeting Proceedings	Marcum JW, Olson SL, Ferkul SV. "Mixed convection blowoff limits as a function of oxygen concentration and upward forced stretch rate for burning PMMA rods of various sizes." 47th International Conference on Environmental Systems, Charleston, SC, July 16-20, 2017. 47th International Conference on Environmental Systems, Charleston, SC, July 16-20, 2017. ICES paper ICES-2017-370. , Jul-2017